

PHILADELPHIA HARBOR, PA, USA

WMO: 997286

Lat: 39.9330N

Lon: 75.1420W

Elev: 20

StdP: 101.08

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.4	(c) -7.2	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.5	(c) 35.3	(d) N/A	(e) 33.9	(f) N/A	(g) 32.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(d) DB	(e) -12.1	(f) 38.1	(g) 2.7	(h) 1.5	(i) -14.1	(j) 39.2	(k) -15.6	(l) 40.1	(m) -17.1	(n) 40.9	(o) -19.1	(p) 42.0	(q) N/A
(4) N/A	(5) N/A	(6) N/A	(7) WB	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A	(19) N/A	(20) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	1.4	2.3	6.7	13.5	19.2	24.2	27.2	25.9	21.9	15.6	9.0	4.8
	DBStd	9.89	5.25	4.88	4.80	4.72	4.17	3.33	2.56	2.67	3.29	4.16	4.15	4.55
	HDD10.0	875	270	222	124	17	0	0	0	0	0	5	66	171
	HDD18.3	2344	526	450	361	158	39	2	0	0	6	104	280	418
	CDD10.0	2478	3	4	23	123	287	427	533	493	358	179	37	11
	CDD18.3	906	0	0	1	14	67	179	275	234	114	21	0	0
	CDH23.3	8309	0	1	6	157	582	1618	2951	2136	754	104	1	0
	CDH26.7	3419	0	0	0	48	200	666	1392	859	233	21	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	1087	80	70	93	90	90	98	109	101	102	76	79	92
	PrecMax	1641	225	164	186	230	179	268	336	491	332	221	230	225
	PrecMin	745	13	14	17	13	12	5	17	12	5	2	8	20
	PrecStd	197	38	34	42	44	40	59	62	73	66	41	51	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	19.1	24.0	31.3	33.7	36.5	38.0	36.3	33.8	28.9	21.6	19.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	14.5	15.1	20.1	27.6	31.1	34.1	35.8	34.2	31.7	26.2	19.4	16.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	11.8	11.7	16.7	24.6	28.9	32.3	34.3	32.6	29.4	23.9	17.0	14.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	9.1	9.0	14.1	21.7	26.8	30.5	32.8	31.1	27.5	21.9	15.3	11.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	6.9	7.4	8.6	9.8	10.0	9.5	9.5	8.7	8.2	7.8	7.2	6.4
		MCDBR	11.2	11.3	13.2	14.4	12.7	11.7	11.3	10.5	10.6	10.2	9.9	9.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.315	0.322	0.364	0.395	0.442	0.504	0.508	0.500	0.413	0.362	0.329	0.311
	taud		2.450	2.421	2.365	2.304	2.205	2.073	2.077	2.085	2.302	2.451	2.507	2.513
	Ebn at Noon		850	896	889	879	842	787	780	773	826	839	829	829
	Edh at Noon		80	94	111	126	143	163	161	156	118	91	76	70
All-Sky Solar Radiation	RadAvg		1.93	2.83	3.75	4.85	5.40	5.91	6.02	5.24	4.33	3.10	2.20	1.68
	RadStd		0.17	0.31	0.30	0.42	0.64	0.48	0.28	0.34	0.47	0.32	0.20	0.17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (74 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+121	+69

Nomenclature: See separate page